3.4 Properties of Exponential Functions – Worksheet

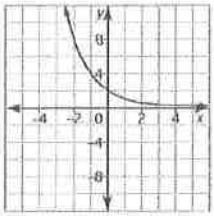
SOLUTIONS

MCR3U

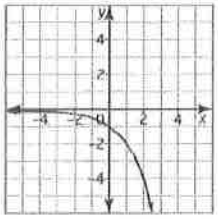
Iensen

1) Match each graph with its corresponding equation.

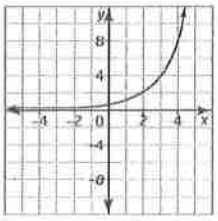
a) B



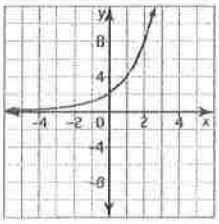
b) D



c) C



d) A



A $y = 2(2)^x$

B $y = 2\left(\frac{1}{2}\right)^x$

C $y = \frac{1}{2}(2)^x$

D $y = -2^x$

2) Given the following exponential equations, state whether they are increasing or decreasing.

a) $y = 3\left(\frac{1}{2}\right)^x$ D

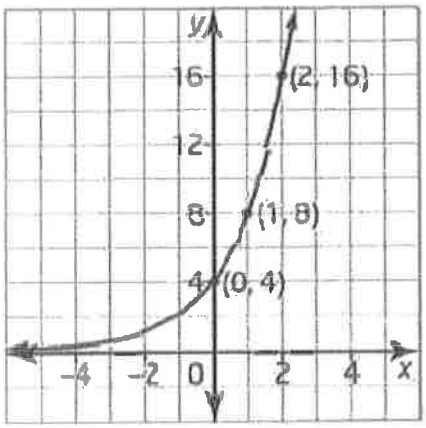
b) $y = -3\left(\frac{1}{2}\right)^x$ I

c) $y = 3(2)^x$ I

d) $y = -3(2)^x$ D

e) $y = 3(2)^{-x}$ $y = 3\left(\frac{1}{2}\right)^x$ D

3) Write an exponential equation to match the graph shown

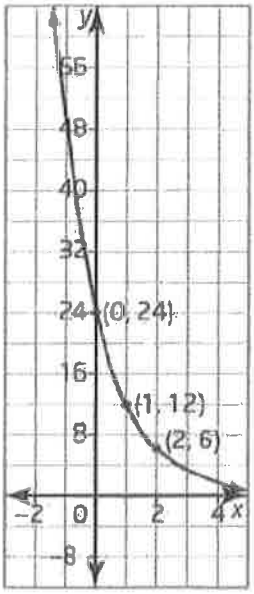


$a = 4$

$b = 2$

$y = 4(2^x)$

4) Write an exponential equation to match the graph shown



$a = 24$

$b = \frac{1}{2}$

$y = 24\left(\frac{1}{2}\right)^x$

5) A radioactive sample with an initial mass of 25 mg has a half-life of 2 days.

a) Write an equation to model this exponential decay where t is the time, in days, and A is the amount of the substance that remains.

$A = 25\left(\frac{1}{2}\right)^{t/2}$

b) What is the amount of radioactive material remaining after 7 days?

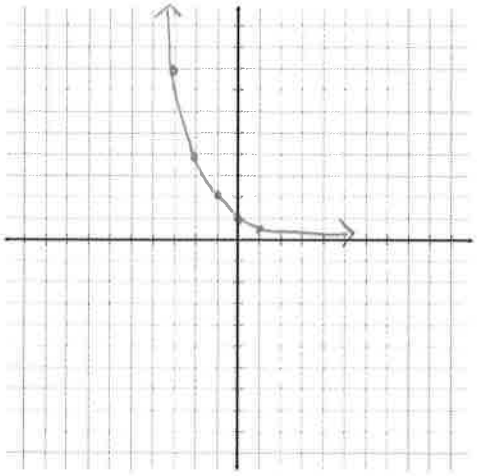
$A = 25\left(\frac{1}{2}\right)^{7/2}$

$A = 2.2 \text{ mg}$

6) Graph each function and identify the...

i) domain
ii) range
iii) x- and y-intercepts, if they exist
iv) increasing or decreasing
v) asymptote

a) $f(x) = \left(\frac{1}{2}\right)^x$



i) $D: \{x \in \mathbb{R}\}$

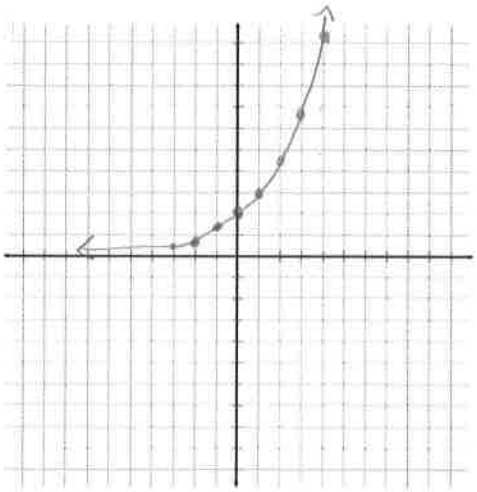
ii) $R: \{y \in \mathbb{R} \mid y > 0\}$

iii) y-intercept at $(0,1)$, no x-intercept.

iv) decreasing

v) $y = 0$ is a horizontal asymptote.

b) $y = 2(1.5^x)$



i) $D: \{x \in \mathbb{R}\}$

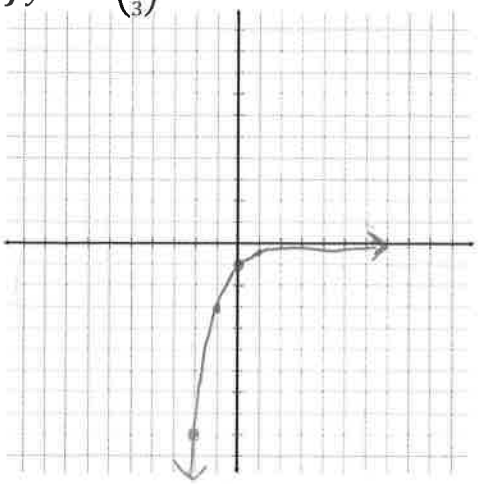
ii) $R: \{y \in \mathbb{R} \mid y > 0\}$

iii) y-int at $(0,2)$, no x-int.

iv) increasing

v) $y = 0$ is a horizontal asymptote.

c) $y = -\left(\frac{1}{3}\right)^x$



i) $D: \{x \in \mathbb{R}\}$

ii) $R: \{y \in \mathbb{R} \mid y < 0\}$

iii) y-int at $(0,-1)$

iv) increasing

v) $y = 0$ is a horizontal asymptote.

Answers

1) a) B **b)** D **c)** C **d)** A

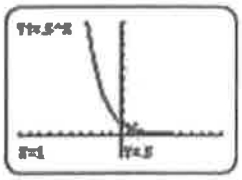
2) a) decreasing **b)** increasing **c)** increasing **d)** decreasing **e)** decreasing

3) $y = 4(2^x)$

4) $y = 24\left(\frac{1}{2}\right)^x$

5) a) $A = 25\left(\frac{1}{2}\right)^{\frac{t}{2}}$ **b)** 2.2 mg

6)

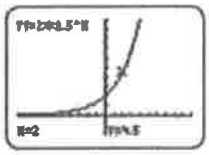
a) 

i) $\{x \in \mathbb{R}\}$

ii) $\{y \in \mathbb{R}, y > 0\}$

iii) no x-intercept; y-intercept 1

iv) always decreasing **v)** $y = 0$

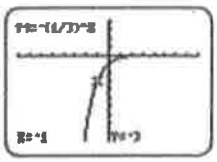
b) 

i) $\{x \in \mathbb{R}\}$

ii) $\{y \in \mathbb{R}, y > 0\}$

iii) no x-intercept; y-intercept 2

iv) always increasing **v)** $y = 0$

c) 

i) $\{x \in \mathbb{R}\}$

ii) $\{y \in \mathbb{R}, y < 0\}$

iii) no x-intercept; y-intercept -1

iv) always increasing **v)** $y = 0$